

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
Sample Paper for Practice / Class Discussion

Course No. : AIMLCZG545
Course Title : Quantum Machine Learning
Nature of Exam : Open Book (As per Course Handout)

Note to Students: This paper is for practice and in-class discussion only – it does not appear in, and is not a preview of, the Regular or Makeup examinations. Solutions are included below each question.

Q1. **[3 Marks]**

A factory dashboard represents the on/off status of three machines on the shop floor as a 3-bit binary flag vector, encoded directly into computational basis states (basis encoding: bit 0 $\rightarrow$ $|0\rangle$, bit 1 $\rightarrow$ $|1\rangle$):

Reading 1: $x = [1, 0, 1]$, Reading 2: $x' = [1, 1, 1]$

- (a) Write $|\psi(x)\rangle$ and $|\psi(x')\rangle$ as 3-qubit computational basis states. **[1 Marks]**
- (b) Compute the quantum kernel $K(x, x') = |\langle\psi(x)|\psi(x')\rangle|^2$. **[1 Marks]**
- (c) Readings x and x' differ in only one machine's status (the second bit). In one sentence, state whether basis encoding can express this "partial similarity", and explain why or why not. **[1 Marks]**

— **SOLUTION** —

(a) $|\psi(x)\rangle = |101\rangle$, $|\psi(x')\rangle = |111\rangle$.

(b) These are distinct computational basis states, so $K(x, x') = |\langle 101|111\rangle|^2 = 0$.

(c) No. Basis encoding gives an *exact-match* kernel: the inner product is 1 only if every bit matches exactly, and 0 for any mismatch, however small. Since x and x' differ in only one out of three bits, the kernel still reports $K = 0$, the same value it would report for two readings that differ in all three bits – basis encoding cannot represent degrees of similarity, only identical vs. not identical.

Q2. **[3 Marks]**

- (a) A classmate says: "*The ansatz $W(\theta)$ is what determines which functions a VQC can represent; the encoding $S(x)$ just decides how the data enters the circuit.*" Is this state-

ment correct? Answer in 2–3 lines.

[1.5

Marks]

- (b) A team measures $\langle Z \rangle$ from a VQC to predict a machine’s remaining useful life in hours (a continuous value). They later reuse the identical circuit to generate synthetic remaining-useful-life samples for data augmentation. In 2–3 lines, state whether the circuit itself needs to change, and describe what does change. [1.5 Marks]

— SOLUTION —

(a) No, the statement has the roles reversed. It is the **encoding** $S(x)$ that determines *which* functions (frequencies) the VQC can represent at all – it fixes the accessible frequency spectrum Ω . The **ansatz** $W(\theta)$ only adjusts the trainable coefficients (amplitude and phase) of the frequencies already made available by the encoding; a deeper or more complex ansatz cannot introduce frequencies the encoding did not generate.

(b) The circuit itself (the parameterised unitary) does not need to change. What changes is the **output**: from a single expectation value $\langle Z \rangle$ (a deterministic prediction) to the full output probability distribution $p_\theta(x)$ over measurement outcomes (used for sampling). Correspondingly, the **learning objective** changes from a prediction-error loss such as MSE to a distribution-matching objective (e.g. maximum likelihood / KL divergence, or an adversarial objective).

Q3.

[4 Marks]

A retail team builds a simple hybrid demand-forecasting model: 64-dimensional PCA-reduced features $\rightarrow$ Q-Mapper $\rightarrow$ a 3-qubit VQC (2 variational layers, 1 R_y per qubit per layer) $\rightarrow$ classical head, trained as a Series pipeline.

- (a) [Numerical] Compute the total number of trainable parameters in the Q-Mapper, the VQC, and the classical head. [2 Marks]
- (b) [Analysis] A colleague suggests replacing the linear Q-Mapper with a small 2-layer nonlinear MLP ”to capture more complex feature interactions before encoding.” In 2–3 lines, evaluate this suggestion, referencing the module’s ablation finding. [2 Marks]

— SOLUTION —

(a)

- Q-Mapper: $n(d + 1) = 3 \times 65 = 195$.
- VQC: layers $\times$ qubits $= 2 \times 3 = 6$.
- Classical head: $n \times 1 + 1 = 4$.
- Total = 205 trainable parameters.

(b)

- The ablation study found that a **nonlinear** Q-Mapper *degrades* performance relative to a linear one.
 - Reason: the encoding gate (R_y) that follows already introduces the necessary nonlinearity via the quantum feature map – an additional nonlinear classical layer before it is redundant, not complementary.
 - **Recommendation:** reject the suggestion; keep the Q-Mapper linear ($\phi = W_e z + b_e$).
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Q4.

[4 Marks]

A 1-qubit angle-encoding QSVM uses $|\phi(w)\rangle = R_y(w)|0\rangle$ and kernel $\kappa(w, w') = \cos^2\left(\frac{w-w'}{2}\right)$, with just two training points $w_1 = \frac{\pi}{2}$ and $w_2 = \frac{5\pi}{6}$.

- (a) *[Numerical]* Compute the 2×2 Gram matrix K for w_1, w_2 . [2 Marks]
- (b) *[Analysis]* Training produces the RKHS function $f(w) = 0.7\kappa(w_1, w) - 0.4\kappa(w_2, w)$. Evaluate $f(w_1)$ directly, and verify this matches $\langle f, \kappa(w_1, \cdot) \rangle_{\mathcal{F}_\kappa}$ computed using your Gram matrix from part (a). [2 Marks]

— SOLUTION —

(a)

$$\kappa(w_1, w_2) = \cos^2\left(\frac{\pi}{6}\right) = 0.75, \quad K = \begin{pmatrix} 1 & 0.75 \\ 0.75 & 1 \end{pmatrix}$$

(b)

- Direct evaluation: $f(w_1) = 0.7(1) - 0.4(0.75) = \boxed{0.4}$.
 - Via the inner product: $\langle f, \kappa(w_1, \cdot) \rangle_{\mathcal{F}_\kappa} = 0.7 K_{11} - 0.4 K_{21} = 0.7(1) - 0.4(0.75) = \boxed{0.4}$.
 - Both match, confirming the reproducing property on a fresh, smaller example.
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Q5.

[4 Marks]

- (a) A classmate says: "Quantum Reinforcement Learning replaces the entire RL algorithm with a quantum algorithm, not just the neural network." Is this correct? Answer in 2–3 lines, referencing specifically what changes and what stays classical when going from DQN to Quantum DQN. [2 Marks]
- (b) For the return $G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1}$, briefly explain (1–2 lines each) how the agent's behaviour differs when γ is close to 0 versus close to 1. [2 Marks]

— SOLUTION —

(a) Incorrect. Only the **function approximator** is replaced – the classical neural network is swapped for a parameterized quantum circuit (PQC). Everything else stays classical: the

agent-environment loop, the replay buffer and target network (for DQN), the Bellman/policy-gradient update rule, and the optimiser – only the component that maps states to Q-values (or action scores) is quantum.

(b)

- $\gamma \rightarrow 0$: future rewards are heavily discounted, so G_t is dominated by the immediate reward r_{t+1} – the agent behaves **myopically**.
- $\gamma \rightarrow 1$: future rewards count almost as much as immediate ones, so the agent becomes **far-sighted**, willing to sacrifice short-term reward for better long-term return.

Q6.

[4 Marks]

A team is choosing between basis encoding and angle encoding for a small genomics dataset with $N = 4$ features, where each feature would need $\tau = 8$ bits of precision if basis-encoded.

- Compute the number of qubits required for basis encoding ($N\tau$) versus angle encoding (N). Which is more NISQ-feasible? [2 Marks]
- Explain what the resulting Gram matrix would look like under basis encoding versus angle encoding for this dataset, and state which encoding would give the model a genuine ability to generalise to new, unseen samples. [2 Marks]

— SOLUTION —

(a)

- Basis encoding: $N\tau = 4 \times 8 = 32$ qubits.
- Angle encoding: $N = 4$ qubits.
- Angle encoding is far more NISQ-feasible – an $8\times$ reduction in qubit count.

(b)

- Basis encoding maps each distinct feature vector to a distinct, mutually orthogonal basis state, so $\kappa(x, x') = \delta_{x, x'}$ – the Gram matrix becomes (close to) the identity matrix. This is the memorisation regime: no ability to generalise.
- Angle encoding produces a smooth, graded kernel where nearby feature values give high similarity – the Gram matrix has meaningful off-diagonal structure, allowing the model to interpolate to unseen samples.
- **Conclusion:** angle encoding is the better choice on *both* counts – more resource-efficient *and* the encoding that actually allows the model to generalise.

Q7.

[4 Marks]

A team compares two circuits for the same $n = 10$ -qubit problem: (1) a standard hardware-

efficient VQC with $L = 3$ variational layers (one trainable rotation per qubit per layer), and (2) a QAOA circuit with $p = 3$ layers solving MaxCut on the same 10-qubit graph.

- (a) Compute the number of trainable parameters for the VQC (using the layers $\times$ qubits formula) and for the QAOA circuit (using the $2p$ formula). [2 Marks]
- (b) Explain, using QAOA's specific structure (parameters shared globally per layer via the problem/mixer Hamiltonians, rather than one rotation per qubit), why QAOA's parameter count does **not** scale with qubit count n – unlike the generic VQC ansatz. Then use the $2K + 1$ parameter-shift formula to compute the number of circuit evaluations needed for one full gradient step for each circuit, and state the resulting resource consequence. [2 Marks]

— SOLUTION —

(a)

- VQC: $L \times n = 3 \times 10 = \boxed{30 \text{ parameters}}$.
- QAOA: $2p = 2 \times 3 = \boxed{6 \text{ parameters}}$.

(b)

- In a hardware-efficient VQC, every qubit gets its own independent trainable rotation in every layer, so the parameter count grows linearly with n . In QAOA, each layer applies the *same* γ_p (problem Hamiltonian) and β_p (mixer Hamiltonian) across the entire circuit – the Hamiltonian, not a per-qubit rotation, defines the circuit structure. So the parameter count depends only on the depth p , not on n .
- Circuit evaluations for one gradient step ($2K + 1$):

$$\text{VQC: } 2(30) + 1 = \boxed{61 \text{ evaluations}}, \quad \text{QAOA: } 2(6) + 1 = \boxed{13 \text{ evaluations}}$$

- **Consequence:** QAOA needs roughly $5\times$ fewer circuit evaluations (and correspondingly less shot budget) per gradient step than the VQC on the same n -qubit problem, and – because its parameter count is fixed by p alone – this advantage grows even further as n increases, since the VQC's cost keeps scaling with n while QAOA's does not.